Corrected FDA Raman Analysis: Important Diagrams and Statistics

Real-data implementation summary for the corrected milk Raman workflow. No synthetic spectra are used. Controls come from the repository tabular workbooks; bacterial spectra come from the experimental workbook; uploaded raw milk samples are read from the supplied text files when present.

Preprocessing sequence: Savitzky-Golay smoothing (window=5, polyorder=2), ALS baseline correction (lambda=1e6, p=0.01, 10 iterations), negative clipping, and L2 normalization. FDA representation: cubic B-splines via scipy splrep/splev, k=3, s=1e-4, evaluated on 400-3500 cm-1 at 1.25 cm-1 spacing.

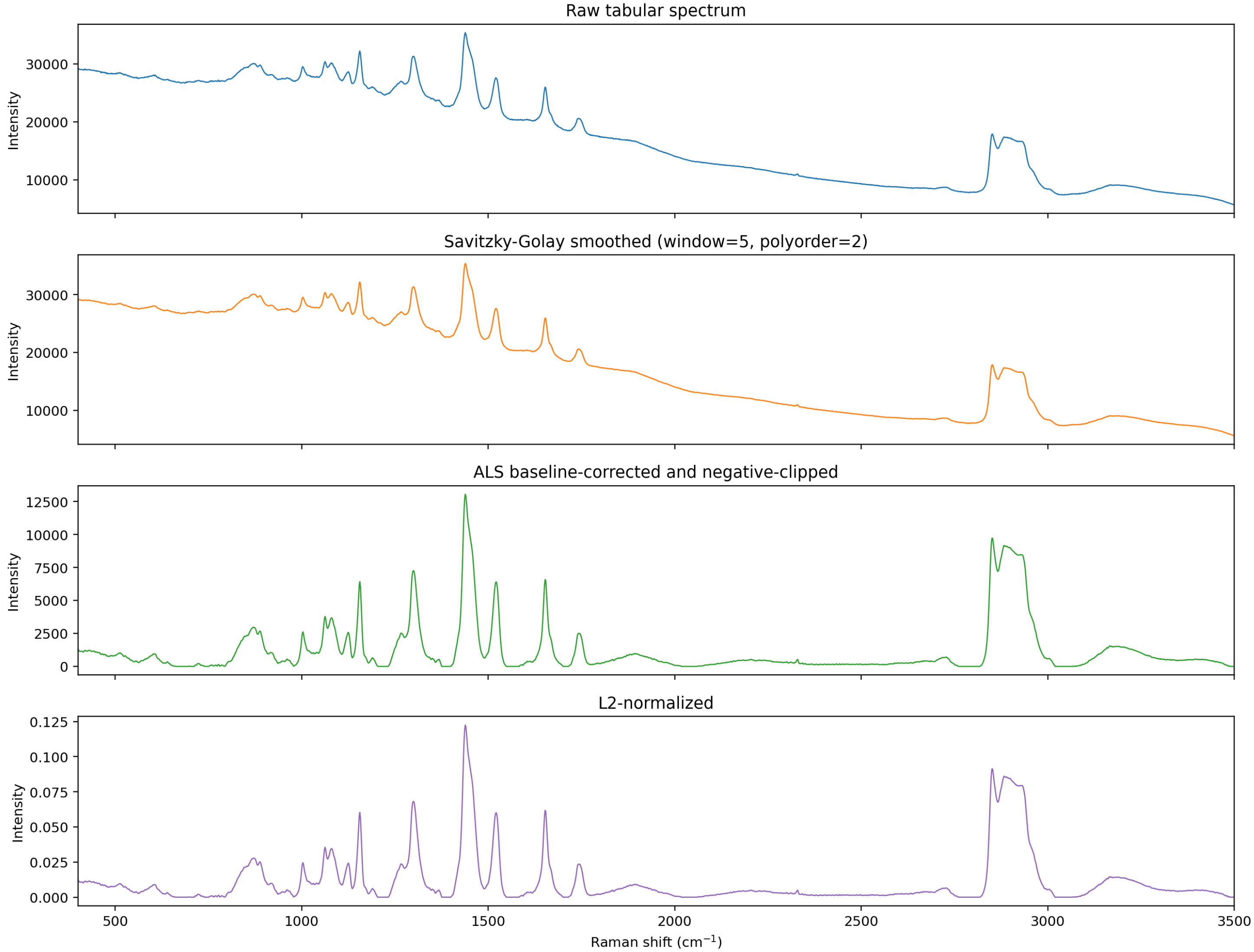
Core Statistics

distance_metric	mean	std	90th_percentile	95th_percentile	99th_percentile	chosen_threshold
l2	0.3996	0.2414	0.7043	0.7635	1.26	0.7635
weighted_l2	0.1943	0.1019	0.3251	0.3598	0.535	0.3598
auc	5.529	3.374	8.781	10.46	14.41	10.46
wasserstein_1d	127	65.8	213.9	241.5	298.4	241.5

sample_group	n	mean_authenticity_score	percent_above_threshold	auroc_combined_score	best_individual_metric	best_individual_metric_auroc
bacterial	9	1.552	100	0.9827	weighted_l2	1
control	45	0.5295	4.444			
raw_unadulterated	5	0.8099	20			

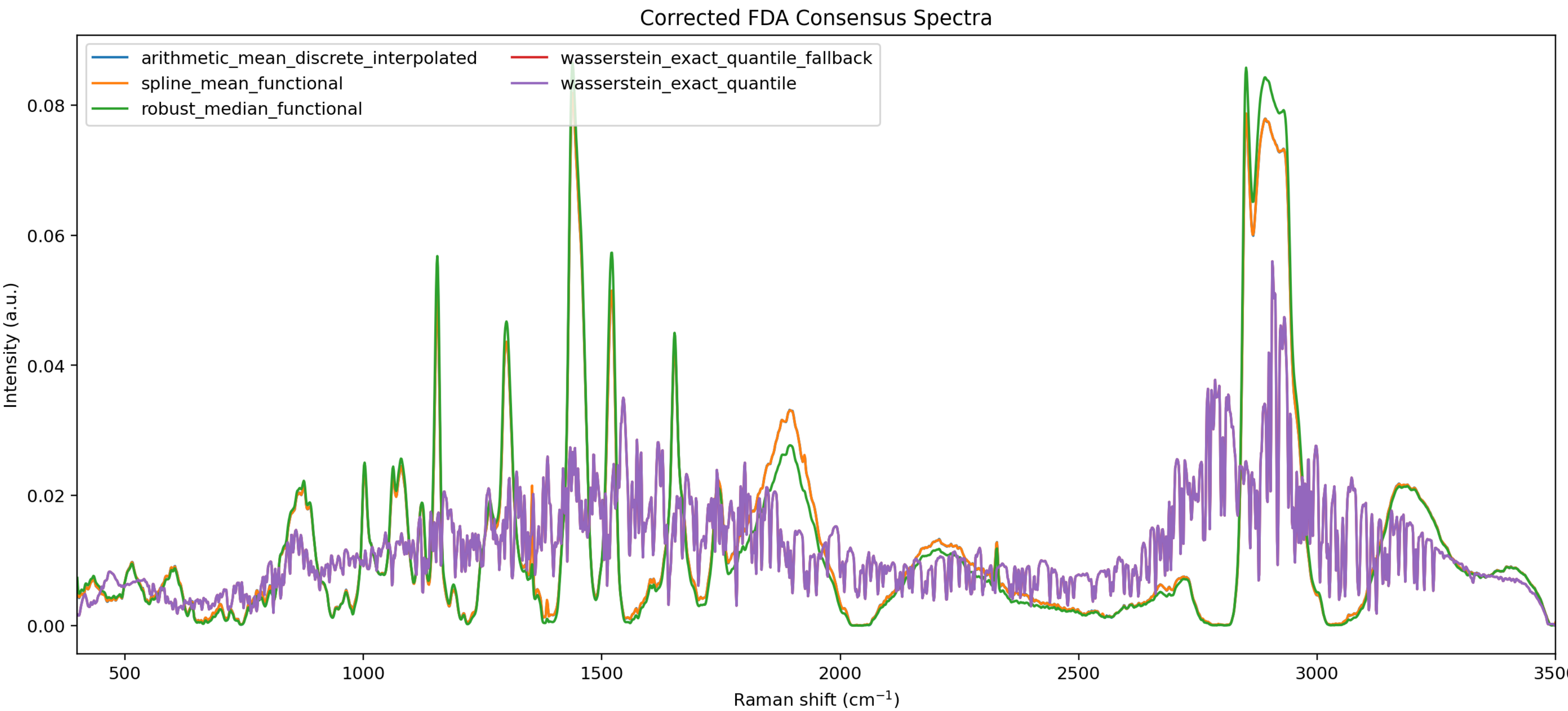
metric	auroc_bacterial_vs_control
weighted_l2	1
authenticity_score	0.9827
l2	0.9778
wasserstein_1d	0.8963
auc	0.7111

Preprocessing Diagnostic



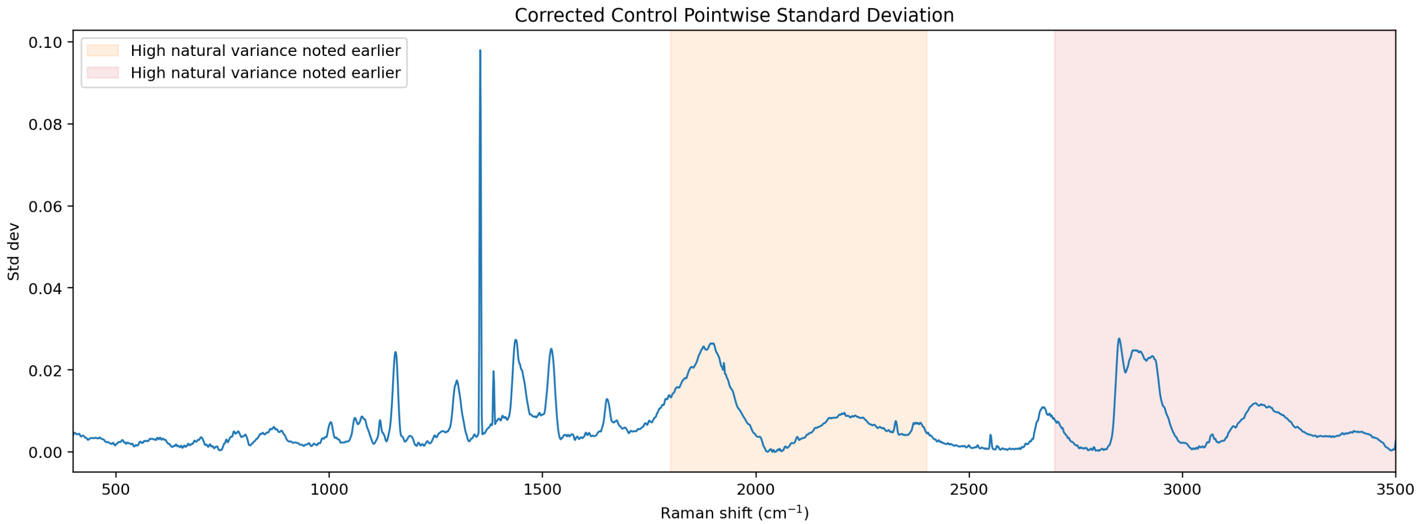
Raw, Savitzky-Golay smoothed, ALS baseline-corrected/clipped, and L2-normalized stages.

Corrected FDA Consensus

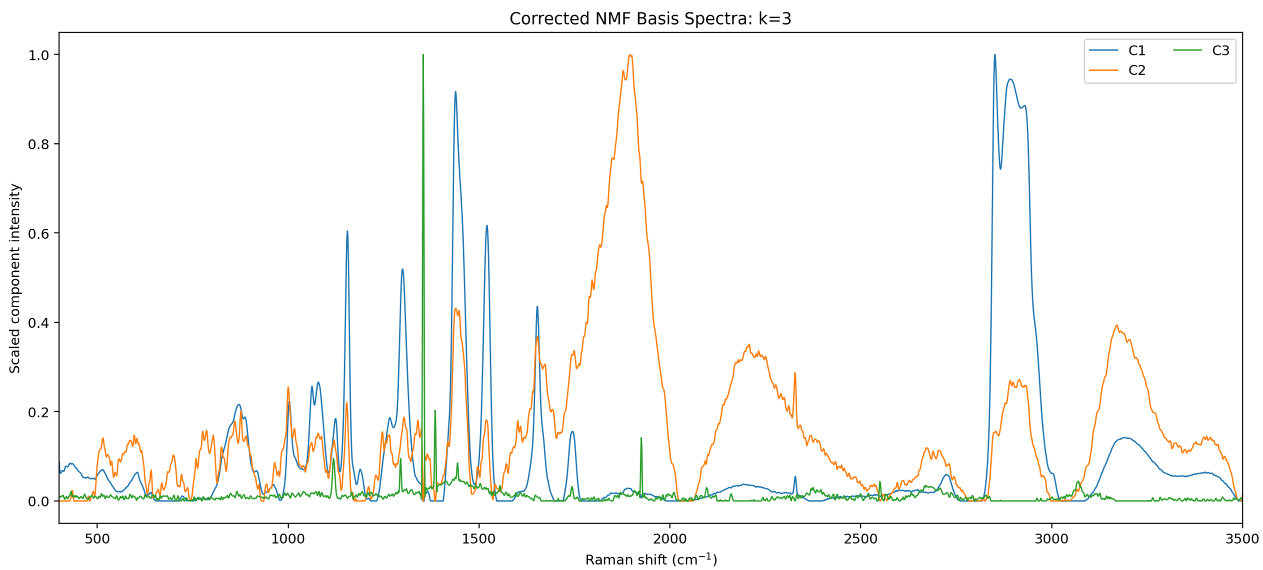


Corrected consensus spectra. The arithmetic mean is discrete/interpolated; spline mean, robust median, and Wasserstein peak locator use FDA-preprocessed spectra.

Variance and NMF Analysis

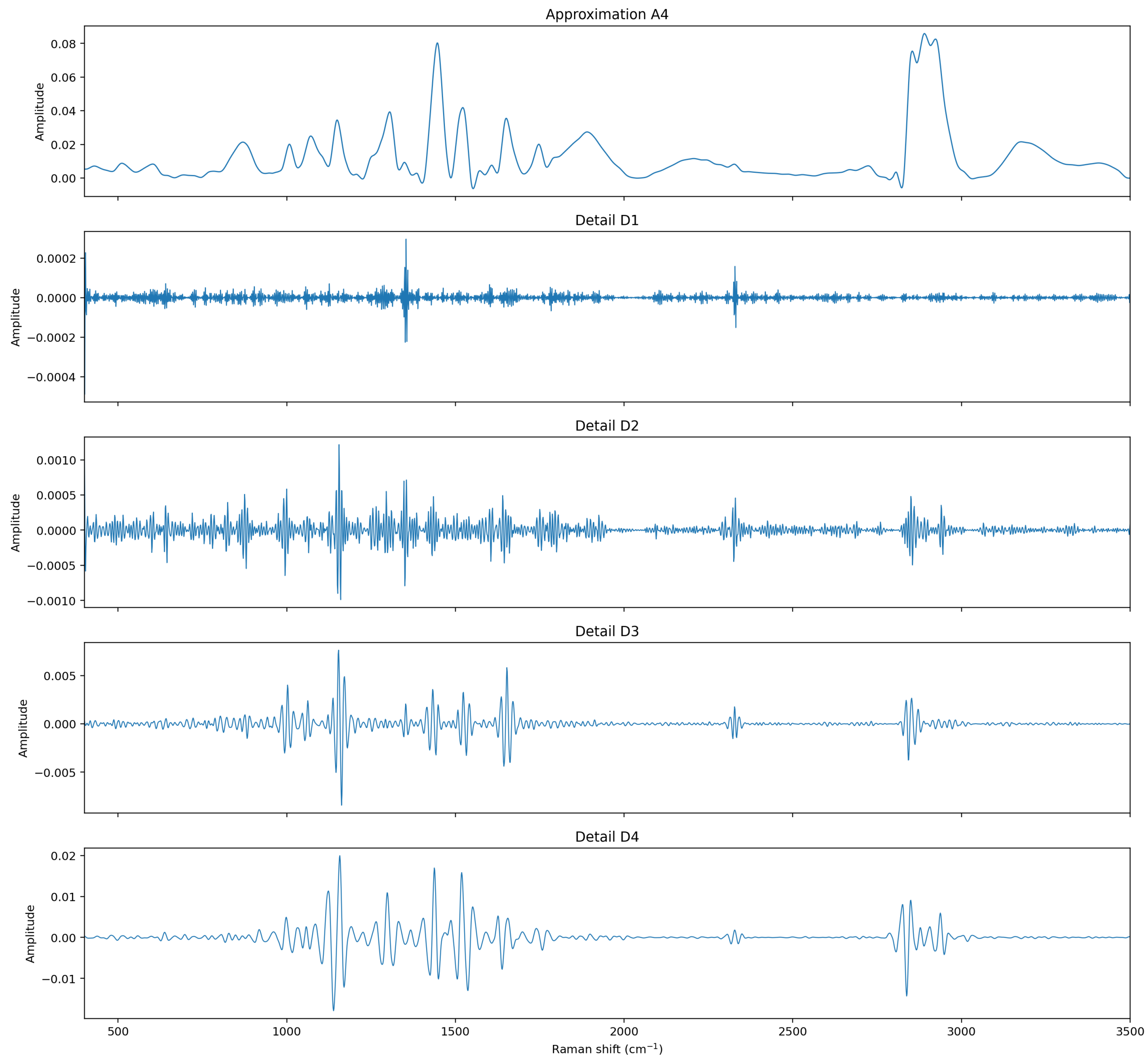


Pointwise standard deviation and variance-weighting profile for corrected control spectra.



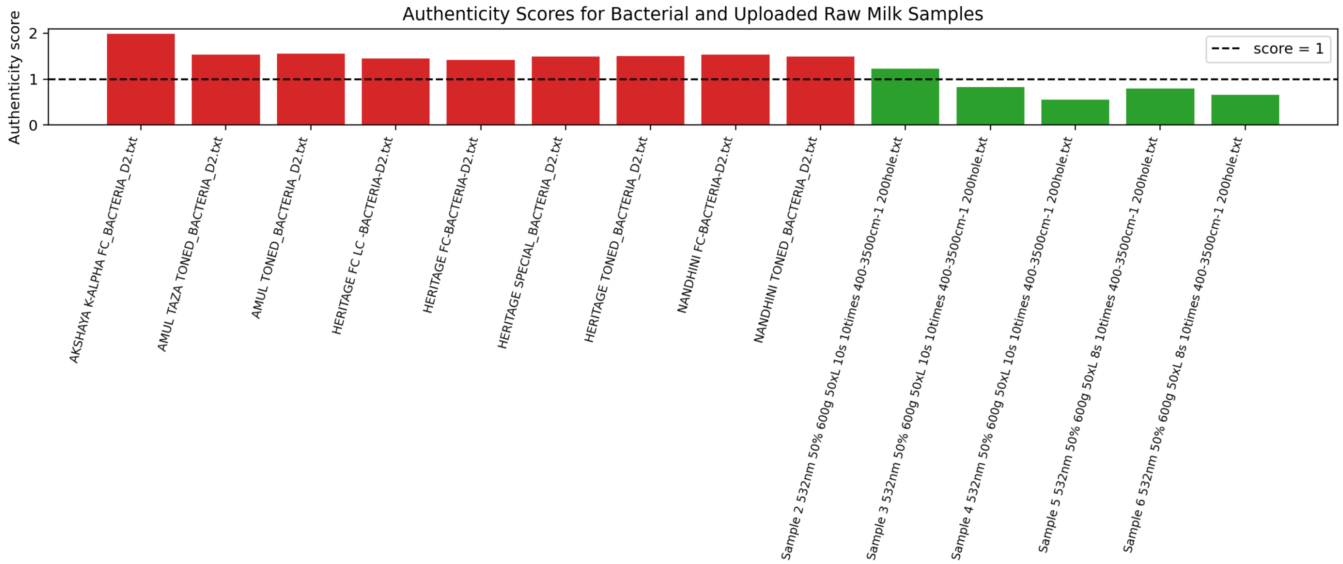
NMF basis spectra for selected k=3, chosen from model-comparison coverage and reconstruction statistics.

Wavelet Analysis

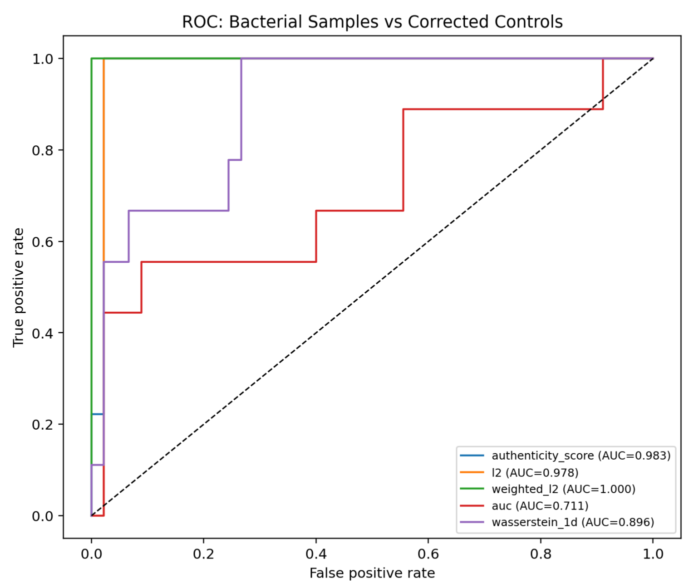


Wavelet decomposition of the corrected robust functional median using sym8 level 4.

Validation Against Bacterial and Uploaded Raw Milk Samples



Authenticity scores for bacterial spectra and uploaded raw unadulterated samples. The dashed line marks score=1.



ROC curves for bacterial-vs-control discrimination using combined and individual distance metrics.

Audit Notes

Low-regularization Sinkhorn trials were attempted and recorded. Because the low-reg audit was numerically unstable/nonconvergent, the pipeline uses the exact 1D quantile Wasserstein fallback as the shared dominant-peak locator.

reg	stable	converged	final_error	iterations	peak_to_median	reason	iteration
0.0001	True	False	0.1794	250	5.943		
1e-05	True	False	0.3424	250	9.744		
1e-06	False					non-finite error	211

Uploaded raw sample limitation: the user described six unadulterated raw milk samples, but only the files below were missing from the local sample directory during execution.

missing_uploaded_raw_sample
Sample 1 532nm 50% 600g 50xL 10s 10times 400-3500cm-1 200hole.txt

Region Definitions and Sources

Fingerprint region: 400-1800 cm-1. CH-stretch region: 2800-3000 cm-1. These choices cover the milk-relevant protein, lipid, carbohydrate, and fatty-acid bands used in the analysis.

Sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC2715834/ ; https://pmc.ncbi.nlm.nih.gov/articles/PMC10052158/ ; https://physicsopenlab.org/2022/01/11/raman-spectroscopy-of-organic-and-inorganic-molecules/ ; https://pubs.acs.org/doi/10.1021/acs.analchem.5c05031